

IQR Assignment

Fig1:

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.334744	66.358558	72.100558	62.278186	277648.648649
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0
25%	54.5	60.6	60.9	61.0	60.0	57.945	240000.0
50%	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
75%	161.5	75.7	73.0	72.0	83.5	66.255	300000.0
100%	215.0	89.4	91.15	88.5	98.0	77.89	390000.0
IQR	107.0	15.1	12.1	11.0	23.5	8.31	60000.0
1.5IQR	160.5	22.65	18.15	16.5	35.25	12.465	90000.0
LesserRange	-106.0	37.95	42.75	44.5	24.75	45.48	150000.0
GreaterRange	322.0	98.35	91.15	88.5	118.75	78.72	390000.0
Min	1	40.89	42.75	50.0	50.0	51.21	200000.0
Max	215	89.4	91.15	88.5	98.0	77.89	390000.0

Info: ref – fig1

The IQR analysis is as follows:

10th Pass Mark (ssc_p):

- 25th percentile (Q1) -> 60.6%
- 75th percentile (Q3) -> 75.7%
- Interquartile Range (IQR) -> 15.1%
- Lesser Outlier -> 37.95%
- Greater Outlier -> 98.35%
- The middle 50% of students scored between 60.6% and 75.7%.
- The spread of the middle 50% of student marks is 15.1%.
- The minimum score (40.89%) is greater than the Lesser Outlier value (37.95%).
- The maximum score (89.4%) is less than the Greater Outlier value (98.35%).
- Therefore, no outliers are present in the 10th Pass Mark data.

HSC Percentage (hsc_p):

- 25th percentile (Q1) -> 60.9%
- 75th percentile (Q3) -> 73.0%
- Interquartile Range (IQR) -> 12.1%
- Lesser Outlier -> 42.75%
- Greater Outlier -> 91.15%
- The minimum score (42.75%) and maximum score (91.15%) lie within the Lesser and Greater Outlier limits.
- Therefore, no outliers are present in the HSC Percentage data.

Degree Percentage (degree_p):

- 25th percentile (Q1) -> 61.0%
- 75th percentile (Q3) -> 72.0%
- Interquartile Range (IQR) -> 11.0%
- Lesser Outlier -> 44.5%
- Greater Outlier -> 88.5%
- The minimum score (50.0%) and maximum score (88.5%) lie within the Lesser and Greater Outlier limits.
- Therefore, no outliers are present in the Degree Percentage data.

Entrance Test Percentage (etest_p):

- 25th percentile (Q1) -> 60.0%
- 75th percentile (Q3) -> 83.5%
- Interquartile Range (IQR) -> 23.5%
- Lesser Outlier -> 24.75%
- Greater Outlier -> 118.75%
- The minimum score (50.0%) and maximum score (98.0%) lie within the Lesser and Greater Outlier limits.
- Therefore, no outliers are present in the Employability Test Percentage data.

MBA Percentage (mba_p):

- 25th percentile (Q1) -> 57.95%
- 75th percentile (Q3) -> 66.26%
- Interquartile Range (IQR) -> 8.31%
- Lesser Outlier -> 45.48%
- Greater Outlier -> 78.72%
- The minimum score (51.21%) and maximum score (77.89%) lie within the Lesser and Greater Outlier limits.
- Therefore, no outliers are present in the MBA Percentage data.

Salary:

- 25th percentile (Q1) -> INR. 240,000
- 75th percentile (Q3) -> INR. 300,000
- Interquartile Range (IQR) -> INR. 60,000
- Lesser Outlier -> INR. 150,000
- Greater Outlier -> INR. 390,000
- The minimum salary (INR. 200,000) is greater than the Lesser Outlier value.
- The maximum salary (INR. 390,000) is equal to the Greater Outlier value.
- Therefore, no outliers are present in the Salary data.